Theoretical study of the He–HCN, Ne–HCN, Ar–HCN, and Kr–HCN complexes

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The two-dimensional potential energy surfaces for the He–HCN, Ne–HCN, Ar–HCN, and Kr–HCN complexes are presented. Calculations have been performed using single and double excitation coupled-cluster theory with noniterative treatment of triple excitations [CCSD(T)] and the augmented correlation-consistent polarized triple-zeta basis set (aug-cc-pVTZ) with an additional $(3s3p2d2f1g)$ set of bond functions. The potentials have been used to find the vibration–rotation energies of the four complexes and their deuterated analogs. The frequencies of rotational or rovibrational transitions found for He–HCN and Ar–HCN are in very good agreement with the experimental results. Good agreement is also obtained with the experimental rotational transition frequencies for Kr–HCN. For Ne–HCN, on the other hand, the agreement with the experimental data is not as good, but can be improved by using larger basis sets. [DOI: 10.1063/1.1332117]

I. INTRODUCTION

Complexes of a rare gas (Rg) atom with the HCN molecule have been the subject of many studies by both experimentalists and theorists.[1–20] The first of these studies was concerned with the microwave spectrum of the Kr–HCN complex,[1] but overall Ar–HCN is certainly the most thoroughly examined member of the Rg–HCN family.[2–9,11,14,15,17–20] A characteristic feature of this complex, in comparison with other weakly bonded systems containing argon such as Ar–HF or Ar–HCl, is its exceptional floppiness.[2] By measuring the rotational spectrum of Ar–HCN Leopold *et al.*[2] showed that the complex should be regarded as a highly nonrigid, easily distortable molecule. Their conclusion was based on the analysis of the rotational spectrum in the ground state, Σ_0, which showed remarkably large centrifugal distortion and was unexpectedly sensitive to isotopic substitution. This unusual behavior was attributed to the strong coupling of the angular and radial degrees of freedom.[2] The original measurements of pure rotational transitions by Leopold *et al.*[2] were later refined and extended by Klots *et al.*,[3] Bumgarner and Blake,[4] and Fraser and Pine.[5] The first direct observation of the pure intermolecular mode by millimeter spectroscopy was made in 1991 by Cooksy *et al.*[9] who detected the lowest Π_1 bending state at 6.07 cm^{-1}. The subsequent measurement of the Σ_1 bending state at 5.50 cm^{-1} was reported two years later by Drucker *et al.*[11] These low-frequency bending modes were shown by Clary *et al.*[6] to arise from the low rotational constant of HCN. In addition to the just summarized experimental studies, several theoretical works concerned with the potential energy surface (PES) for Ar–HCN have also been published.[6,14,19] The most recent of them[19] shows the presence of three minima separated by low barriers in agreement with the early prediction by Leopold *et al.*[2]

In comparison with Ar–HCN, the number of publications concerned with other Rg–HCN complexes is much smaller. Apart from the already mentioned work on Kr–HCN,[1] only one study of Ne–HCN,[12] and two of He–HCN,[13,16] have appeared so far. In 1993, Gutowsky *et al.*[12] observed a few rotational transitions for eight isotopic species of Ne–HCN at low J (0–3). They reported the results for $K=0$ and $K\neq 0$, noting, in the latter case, the difficulty in distinguishing doublets of a vibrational origin from doublets of a rotational origin. In the end, the measured transitions and the energy levels involved were labeled in the standard rotational asymmetric top notation. This choice was somewhat surprising considering the fact that Leopold *et al.*[2] in their analysis of the rotational spectrum of Ar–HCN also raised the question whether the complex should be treated as an asymmetric rotor or as a linear rotor with large centrifugal distortion. They concluded that no asymmetric rotor structure could account for the observed spectrum.[2]

From an experimental point of view the He–HCN complex, due to its weak binding, is more difficult to deal with than other members of the Rg–HCN family. Nonetheless, Drucker *et al.*[13] reported a millimeter wave and microwave spectra that allowed for assignment of majority of the bound energy levels. They also developed the first *ab initio* PES using fourth-order Møller–Plesset perturbation theory (MP4) and a $[7s5p3d/5s3p]$ basis set with an additional $(3s3p)$ set of bond functions. Using this surface as a starting point and the eight available experimental data, Atkins and Hutson[16] developed two empirical potential energy surfaces based on two different functional forms. Even though the two potentials could reproduce the experimental data with similar accuracy, their shapes away from the linear He–HCN geometry were significantly different.

Initially, we intended to devote this work solely to the Ne–HCN complex with the main goal of developing the lacking *ab initio* surface. However, we found that agreement between our calculated rotational transition frequencies and experimental results of Gutowsky *et al.*[12] was not particularly good, even though in our studies of other complexes in which a similar methodology was used the agreement with experiment was much better.[21–23] We therefore decided to examine also the He–HCN complex and develop a new *ab initio* PES for it. The new surface was found to be remarkably similar to one of the empirical potentials of Atkins and Hutson.[16] The calculated transitions were found to be in very good agreement with the transitions observed by Drucker *et al.*[13] At this stage we decided to consider also Ar–HCN and Kr–HCN turning our study into a comparative investigation of the Rg–HCN family. By examining four related complexes using the same methodology and in the same work we think we can show more convincingly the capabilities and limitations of the state-of-the-art *ab initio* methods. Our earlier study of rare gas dimers where we also considered several related complexes, allowed us to see certain trends which, had we focused on only one of them, would have been impossible to see.[24]

The plan of this work is as follows. The next section contains a brief description of our methodology. In Sec. III we describe the potential energy surfaces for the four complexes and compare the calculated vibration–rotation energies with the available experimental and theoretical data. In Sec. IV we discuss the limitations of our approach and report predictions of energies and other quantities for vibration–rotation states that have not yet been observed. Finally, conclusions are presented in Sec. V.

II. METHODOLOGY

A. *Ab initio* calculations

The interaction energy at each geometry examined was found using the supermolecule approach. Calculations, in which only the valence electrons were correlated, were performed using the single and double excitation coupled-cluster approach with triple excitations treated in a noniterative way [CCSD(T)].[25] For all atoms the augmented correlation-consistent triple-zeta (aug-cc-pVTZ)[26–29] basis sets were used. They were supplemented with a $(3s3p2d2f1g)$ (Ref. 24) set of bond functions centered in the midpoint between the center of mass of HCN and a rare gas atom. In the discussion that follows the overall basis set is denoted as avtz+(33221). By using the CCSD(T) approach and the avtz+(33221) basis sets we obtained high quality results for rare gas dimers[24] and Ar–CO.[21] For certain selected geometries of each complex we performed additional calculations using the augmented correlation-consistent quadruple-zeta (aug-cc-pVQZ)[26–29] basis sets with bond functions $(3s3p2d2f1g)$. The overall basis set is denoted as avqz+(33221). For reasons that will be explained later in the article, we decided to develop the entire surface for Ne–HCN using the avqz+(33221) basis set which contained 331 contracted Gaussian functions. To correct for basis set superposition error (BSSE), the counterpoise procedure of Boys and Bernardi[30] was used. All calculations were performed using the MOLPRO 2000 package.[31]

The HCN molecule is treated as a linear rigid rotor with the intramolecular distances fixed at the equilibrium values given by Strey and Mills:[32] $r_{CH}=2.01350\,a_0$ and $r_{CN}=2.17923\,a_0$. The intermolecular geometries are expressed in terms of the standard Jacobi coordinates (R,θ), where R denotes the distance between the center-of-mass of HCN and a rare gas atom, and θ denotes the angle between the HCN molecular axis and the direction of the $\mathbf{R}$ vector. $\theta=0°$ corresponds to the linear Rg–HCN geometry.

In addition to surfaces for the Rg–HCN complexes we also developed surfaces for deuterated complexes. They were simply found by calculating the Jacobi coordinates with respect to the center-of-mass of DCN instead of HCN. The same procedure can also be used to find the surfaces for any other isotopomer of HCN.

B. The potential form

The calculated CCSD(T) interaction energies for each complex were fitted to the functional form proposed for atom-linear molecule systems by Bukowski *et al.*[33] This function was previously tested and shown to provide reliable results for a number of complexes consisting of a rare gas atom and either a closed-shell[19,34,21] or an open-shell[35,22,23] molecule. It contains a short range, V_{sh}, and an asymptotic, V_{as}, components

$$V(R,\theta)=V_{sh}(R,\theta)+V_{as}(R,\theta), \tag{1}$$

where

$$V_{sh}(R,\theta)=G(R,\theta)e^{D(\theta)+B(\theta)R}, \tag{2}$$

and

$$V_{as}(R,\theta)=f_6(|B(\theta)R|)\frac{c_0P_0(\cos\theta)+c_2P_2(\cos\theta)}{R^6}+f_7(|B(\theta)R|)\frac{c_1P_1(\cos\theta)+c_3P_3(\cos\theta)}{R^7}. \tag{3}$$

In Eq. (2) $D(\theta)$, $B(\theta)$, and $G(R,\theta)$ denote expansions in Legendre polynomials:

$$X(\theta)=\sum_{l=0}^{5}x_lP_l(\cos\theta), \quad (X=B,D, \quad \text{and} \quad x=b,d), \tag{4}$$

and

$$G(R,\theta)=\sum_{l=0}^{5}(g_{0,l}+g_{1,l}R+g_{2,l}R^2+g_{3,l}R^3)P_l(\cos\theta). \tag{5}$$

V_{as} is damped with the Tang–Toennies[36] damping function, $f_n(x)$, defined by

$$f_n(x)=1-e^{-x}\sum_{k=0}^{n}\frac{x^k}{k!}. \tag{6}$$

Overall, the potential for each complex was found by optimizing in nonlinear least-squares fits 40 adjustable parameters: b_l, c_l, d_l, and $g_{m,l}(m=0-3)$.

TABLE I. The CCSD(T) interaction energies (in μE_h) obtained with the avtz+(33221) basis set for He–HCN.

R [a_0]	θ=0°	20°	40°	60°	80°	90°	100°	120°	140°	160°	180°
4.50				7383.1	4465.3	4203.2	4430.4	6966.7	14287.1		
5.00			8251.8	2525.8	1427.9	1323.6	1396.6	2325.3	5069.4	9469.1	
5.50			2741.6	701.7	339.9	304.7	325.8	639.2	1609.6	3174.9	4050.3
6.00	5755.0	3243.5	723.4	84.3	−8.8	−17.6	−13.0	78.4	393.6	921.2	1218.4
6.50	1582.0	814.1	66.3	−87.9	−94.5	−94.1	−94.7	−77.2	8.9	169.4	262.8
7.00	266.0	66.1	−104.6	−111.3	−96.2	−94.4	−95.1	−99.1	−86.4	−48.7	−24.3
7.50	−83.1	−114.9	−120.7	−93.5	−77.1	−74.7	−75.8	−83.9	−91.2	−90.3	−87.3
8.00	−135.9	−126.7	−97.7	−69.9	−56.7	−55.1	−55.9	−63.5	−73.5	−81.2	−83.6
8.50	−114.4	−99.7	−71.2	−50.2	−40.8	−39.6	−40.2	−45.8	−54.6	−62.4	−65.5
9.00	−83.1	−71.3	−50.3	−35.7	−29.2	−28.4	−28.8	−32.8	−39.4	−45.6	−48.1
10.00	−40.0	−34.6	−25.1	−18.3	−15.3	−14.9	−15.1	−17.1	−20.4	−23.6	−24.9
12.00	−10.8	−9.7	−7.5	−5.7	−4.9	−4.7	−4.8	−5.3	−6.2	−7.1	−7.4

	0°	105°	110°	115°	120°	180°
6.70		−100.0	−99.6	−98.0		
6.80		−100.0	−100.4	−99.9		
6.90		−99.5	−99.6	−100.0	−99.4	
7.00				−98.7		
7.60						−89.3
7.70						−89.6
7.80						−88.5
7.90	−135.1					
8.10	−134.3					

C. Calculation of vibration–rotation energies

All calculations of the nuclear motion were performed using the BOUND program.[37] The Schrödinger equation with the Hamiltonian of the form[38]

$$H=-\frac{\hbar^2}{2\mu}R^{-1}\frac{d^2}{dR^2}R+H_{\mathrm{int}}+V(R,\theta), \tag{7}$$

was solved with the coupled channel method.[38] The basis set was determined by the maximum rotational quantum number for HCN, $j=30$. The rotational constants, B_e, of HCN and DCN were equal to, respectively, 1.478 221 834 cm^{-1} and 1.207 750 953 cm^{-1}.[39] The reduced masses of the complexes were calculated using either the mass of hydrogen or deuterium and the masses of the most common isotopes of a rare gas atom, carbon, and nitrogen. They were as follows: 3.486 027 184 m_u for He–HCN, 3.502 261 477 m_u for He–DCN, 11.488 837 20 m_u for Ne–HCN, 11.667 071 79 m_u for Ne–DCN, 16.117 171 91 m_u for Ar–HCN, 16.470 143 93 m_u for Ar–DCN, 20.433 430 22 m_u for Kr–HCN, and 21.004 119 62 m_u for Kr–DCN. The coupled equations were solved using the modified log–derivative method of Manolopoulos.[40] The grid for the R coordinate was determined by $R_{\mathrm{min}}=2$ Å, $R_{\mathrm{max}}=18$ Å, and a step size of 0.01 Å.

III. RESULTS

A. Potential energy surfaces

The CCSD(T) interaction energies calculated with the avtz+(33221) basis sets for the He–HCN, Ne–HCN, Ar–HCN, and Kr–HCN complexes are collected in Tables I–IV. The upper part of each table contains the results for the regular grid and the lower part contains the results for a few additional geometries in the vicinity of the minima. The corresponding contour plots are shown in Fig. 1. All four surfaces are qualitatively similar, although, of course, there are some important quantitative differences between them. As expected, the weakest interaction is found for He–HCN and the strongest for Kr–HCN. The global minima are at linear Rg–HCN configurations, $\theta=0°$, with additional shallower minima at bent geometries with $\theta\approx50°$–60° and $\theta\approx100°$–110°. The linear Rg–NCH geometry, $\theta=180°$, can represent either another local minimum or a saddle point. We note that the area of each surface in the vicinity of the local minima at bent geometries is very flat and therefore difficult to characterize reliably. Further increase of basis set size or the use of a more sophisticated method can affect not only the depth of these minima, but their number. We also note the importance of choosing a sufficiently dense grid, as otherwise the minima at bent geometries and very low barriers between them can easily be overlooked.

The global minimum on the avtz+(33221) PES of He–HCN is in the linear He–HCN configuration at $R=7.97\,a_0$ and has a depth of 29.90 cm^{-1} (136.2 μE_h). There is also a local minimum at a bent geometry with a depth of 22.07 cm^{-1} (100.6 μE_h) at $R=6.78\,a_0$ and $\theta=108.5°$. By comparison, Drucker *et al.*[13] found on their MP4 PES only one minimum in the linear He–HCN configuration whose depth was 4.6 cm^{-1} (21 μE_h) shallower than ours.

The avtz+(33221) potential energy surface for Ne–HCN contains four minima. As with He–HCN, the global minimum is in the linear Ne–HCN configuration at $R=8.03\,a_0$ and has a depth of 56.97 cm^{-1} (259.6 μE_h). The two local minima at bent configurations have well depths of 55.55 cm^{-1} (253.1 μE_h) at $R=7.05\,a_0$ and $\theta=51.2°$, and 53.46 cm^{-1} (243.6 μE_h) at $R=6.67\,a_0$ and $\theta=106.6°$. The shallowest minimum with a depth of 48.79 cm^{-1} (222.3 μE_h) occurs at $R=7.57\,a_0$ in the linear Ne–NCH configuration.

TABLE II. The CCSD(T) interaction energies (in μE_h) obtained with the avtz+(33221) basis set for Ne–HCN.

R [a_0]	θ=0°	20°	40°	60°	80°	90°	100°	120°	140°	160°	180°
4.50					8979.2	8505.2	9190.0	15 236.8			
5.00				5195.1	2728.0	2533.9	2751.4	4897.2	10911.3		
5.50			5976.1	1387.6	562.7	495.2	557.1	1259.9	3332.4	6649.1	8511.0
6.00	13 092.4	7297.3	1597.2	125.9	−96.7	−114.1	−101.1	94.7	746.8	1836.2	2453.7
6.50	3663.0	1887.3	166.1	−212.4	−237.6	−238.4	−239.2	−207.9	−40.8	276.9	463.3
7.00	684.2	208.6	−209.6	−247.3	−221.4	−218.4	−222.4	−236.2	−219.4	−154.0	−111.5
7.50	−121.6	−209.9	−247.6	−201.5	−170.4	−167.5	−171.3	−193.3	−215.5	−222.0	−221.4
8.00	−259.3	−248.4	−199.5	−147.1	−123.9	−121.5	−124.6	−142.4	−170.1	−190.9	−198.9
8.50	−224.1	−196.9	−143.6	−104.1	−86.6	−85.4	−88.0	−102.5	−123.7	−143.6	−151.5
9.00	−163.9	−141.2	−101.4	−72.6	−60.4	−59.8	−62.0	−72.0	−87.3	−102.7	−108.9
10.00	−81.2	−70.0	−51.0	−37.6	−31.8	−31.3	−32.2	−35.5	−43.6	−50.6	−53.6
12.00	−24.6	−22.0	−16.8	−12.4	−10.0	−9.4	−9.2	−9.8	−11.2	−12.7	−13.3
	0.0°	45.0°	50.0°	55.0°	60.0°	105.0°	107.5°	110.0°	115.0°	180.0°	
6.50						−237.7		−233.3	−224.0		
6.55						−240.8					
6.60						−242.6	−242.1	−240.8			
6.65						−243.3	−243.2	−242.7			
6.70						−242.9	−243.2	−243.3			
6.75							−242.4	−242.9			
6.80					−250.5						
6.90				−251.8	−250.8						
7.00		−239.3	−251.3	−252.2							
7.10		−249.0	−252.9	−249.0	−241.0						
7.20		−252.7	−250.3								
7.30		−251.0									
7.40										−215.6	
7.60										−222.1	
7.70										−219.4	
7.90	−254.2										
8.10	−258.6										

The avtz+(33221) PES for Ar–HCN contains three minima. The global one is found in the linear Ar–HCN configuration at $R=8.50\,a_0$ and has a depth of 147.0 cm^{-1} (669.7 μE_h) which is approximately 6 cm^{-1} (27.5 μE_h) deeper than the recent CCSD(T) potential calculated with the avtz+(332) basis set[19] and 11 cm^{-1} (50.5 μE_h) deeper than the MP4 potential of Tao *et al.*[14] As with He–HCN, the linear Ar–NCH configuration is a saddle point, but there are two minima at bent geometries. The deeper of them at $R=7.40\,a_0$ and θ=58.4° has a depth of 132.12 cm^{-1} (601.99 μE_h). The shallowest minimum with a depth of 124.76 cm^{-1} (568.46 μE_h) is found at $R=7.11\,a_0$ and θ=103.9°.

The PES for Kr–HCN closely resembles the potential for Ar–HCN as it also contains three minima, although even the shallowest of them is deeper than the global minimum on the Ar–HCN surface. The minima occur at longer intermolecular distances and in comparison with Ar–HCN all three are further away from the center-of-mass of HCN by approximately $0.25\,a_0$. The global minimum is in the linear Kr–HCN configuration at $R=8.76\,a_0$ and has a depth of 179.0 cm^{-1} (815.7 μE_h). The local minima are at $R=7.64\,a_0$ and θ=60.0° and $R=7.38\,a_0$ and θ=104.9°, and have depths 157.2 cm^{-1} (716.1 μE_h) and 148.2 cm^{-1} (675.1 μE_h), respectively.

B. Vibration–rotation energies

1. He–HCN

The He–HCN complex is from an experimental point of view more challenging to study than complexes of heavier rare gases. The formation of the complex is difficult and this is at least partially responsible for the fact that its spectrum was observed later than the spectra of other members of the Rg–HCN family. From a theoretical point of view, however, He–HCN is somewhat more convenient to study because its smaller size makes *ab initio* calculations easier and its shallow potential supports only small number of bound-state energy levels. Our potential supports 19 levels overall, the highest of which corresponds to $J=5$. By comparison, for Ne–HCN and Ar–HCN we find, just for $J=0$, 12, and 35 levels, respectively.

Atkins and Hutson in their study[16] of He–HCN determined two PESs by least-square fitting of two different functional forms to computational and spectroscopic data of Drucker *et al.*[13] The resulting potentials were labeled 1E8 and 2E8, where E8 indicated the fitting to the eight available experimental data. Both 1E8 and 2E8 potentials were found to have significantly deeper wells than the *ab initio* PES of Drucker *et al.*[13] on which they were based. Still, however, their depths of 29.47 cm^{-1} for 1E8 and 29.36 cm^{-1} for 2E8

TABLE III. The CCSD(T) interaction energies (in μE_h) obtained with the avtz+(33221) basis set for Ar–HCN.

R [a_0]	θ=0°	20°	40°	60°	80°	90°	100°	120°	140°	160°	180°
5.00					13 296.3	12 669.6	13 566.2				
5.50				8192.8	4201.3	3923.2	4239.0	7091.8			
6.00			9684.6	2247.9	780.0	684.1	787.6	1807.4	4728.6	9283.5	
6.50		10 758.0	2751.7	98.4	−333.3	−351.9	−323.4	−17.2	987.7	2639.5	3561.2
7.00	5216.5	2908.0	240.4	−524.3	−576.3	−567.9	−564.2	−512.8	−235.5	293.2	601.2
7.50	851.7	195.1	−498.3	−593.8	−531.2	−516.9	−519.4	−546.2	−519.5	−397.9	−316.6
8.00	−443.1	−551.4	−595.6	−498.1	−418.3	−404.8	−408.2	−449.2	−494.0	−503.9	−498.6
8.50	−668.8	−625.9	−502.9	−378.4	−309.4	−299.1	−301.9	−338.8	−392.9	−434.8	−448.9
9.00	−583.2	−515.2	−380.7	−275.6	−223.2	−215.9	−218.7	−247.0	−292.8	−333.8	−349.6
10.00	−315.5	−274.0	−198.1	−142.7	−117.0	−113.4	−114.9	−129.2	−155.0	−178.5	−188.4
12.00	−83.1	−74.2	−56.8	−43.1	−36.5	−35.6	−36.2	−40.5	−47.5	−54.0	−56.9
15.00	−17.8	−16.5	−13.5	−10.7	−9.5	−9.3	−9.5	−9.5	−10.8	−12.0	−12.6
	0°	45°	50°	55°	60°	90°	95°	100°	105°	180°	
7.00							−565.9	−564.2	−560.8		
7.10						−569.8	−568.2	−567.7	−567.4		
7.20						−564.0	−562.8	−563.4	−565.4		
7.30				−584.5	−599.3						
7.40			−575.3	−598.0	−600.5						
7.50			−593.5	−600.8							
7.60		−586.4	−599.6	−595.7							
7.70			−596.8								
7.90										−490.3	
8.10										−498.5	
8.20										−492.5	
8.40	−662.1										
8.60	−664.2										

TABLE IV. The CCSD(T) interaction energies (in μE_h) obtained with the avtz+(33221) basis set for Kr–HCN.

R [a_0]	θ=0°	20°	40°	60°	80°	90°	100°	120°	140°	160°	180°
5.50					7956.8	7491.0	7988.3	12 307.2			
6.00				4815.5	2139.7	1952.4	2127.7	3804.6	8413.5		
6.50			5789.8	947.0	9.3	−44.3	12.3	585.6	2323.6	5076.9	6590.4
7.00	10 022.2	6097.6	1268.0	−386.6	−616.4	−616.5	−601.6	−454.9	110.7	1093.8	1648.3
7.50	2477.7	1209.3	−304.1	−705.6	−682.8	−666.2	−664.7	−666.2	−542.5	−251.2	−75.0
8.00	−93.2	−399.9	−697.1	−668.1	−577.0	−557.4	−559.7	−601.9	−625.7	−584.7	−548.3
8.50	−761.7	−761.0	−676.1	−535.9	−442.0	−426.5	−428.6	−473.9	−534.3	−569.6	−577.4
9.00	−786.7	−708.6	−544.3	−402.2	−327.3	−314.0	−315.8	−353.6	−412.8	−460.6	−478.2
10.00	−472.1	−410.3	−297.4	−213.2	−172.9	−166.5	−167.8	−188.1	−224.1	−256.7	−269.8
12.00	−128.7	−113.9	−86.3	−64.4	−53.6	−50.9	−52.5	−58.7	−68.9	−78.5	−82.5
16.00	−17.1	−15.7	−12.8	−10.2	−8.6	−7.7	−8.5	−9.5	−10.8	−12.0	−12.6
	0°	55°	60°	65°	100°	105°	110°	180°			
7.20					−663.4						
7.30					−673.2	−671.4	−666.6				
7.40				−704.5	−673.0	−673.8	−673.4				
7.50		−674.9		−712.9		−667.8	−670.7				
7.60		−700.6	−714.5	−710.9							
7.70		−712.6	−713.0	−701.1							
7.80		−713.3									
7.90		−705.7									
8.20								−588.6			
8.30								−592.3			
8.40								−587.9			
8.70	−812.4										
8.80	−814.1										
8.90	−804.6										

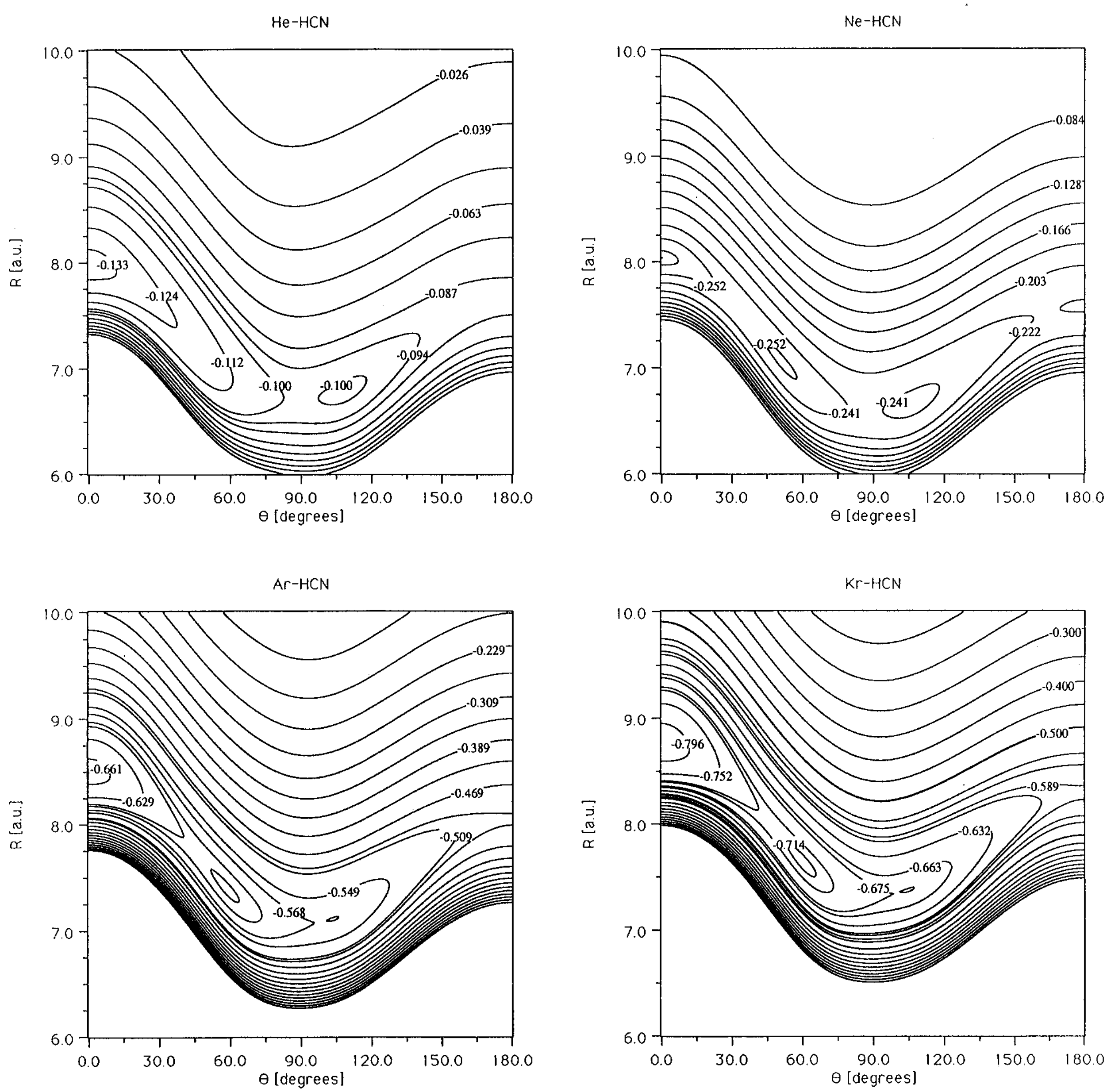

FIG. 1. Contour plots of the potential energy surfaces for the He–HCN, Ne–HCN, Ar–HCN, and Kr–HCN complexes. Contours are labeled in mE_h.

are slightly smaller than the depth of our potential which is 29.90 cm^{-1}.

The 1E8 and 2E8 PESs both have global minima in linear or near-linear He–HCN geometries in agreement with our potential, but the 1E8 surface does not contain the local minimum at a bent geometry in contrast to the 2E8 and our CCSD(T)/avtz+(33221) PESs. In addition to the qualitative similarity, we also find very good agreement between the energies of bound vibration–rotation states calculated using the latter two potentials. Interestingly, the agreement is much better than between the two different E8 potentials as can be seen in Table V which contains the list of energies of bound levels calculated using the indicated PES. The levels are labeled using three quantum numbers, (j,L,J), where j is the rotational quantum number of the HCN molecule, L is the quantum number for rotation of He and HCN about their center-of-mass, and J is the total angular momentum of the complex. This particular convention is known as case 1 coupling scheme[41] and is the most convenient one for weakly anisotropic atom–molecule complexes such as He–HCN.

The energies of vibration–rotation states calculated using our *ab initio* PES are within 1% of the energies obtained with the 2E8 potential, with the exception of the (1,4,4) state which is very close to the dissociation limit where even small differences can result in large percentage errors. We also note the lack of the excited stretching (0,0,0) state on the 2E8 surface, but this level is also very close to the dissociation limit, and additional measurements or more sophisticated calculations will have to be made to verify that a bound state like this actually exists.

In Table VI we compare the calculated transition frequencies to the five spectroscopic data and several predictions made by Atkins and Hutson.[16] As can be seen, the overall agreement is very good, with the exception of the low

TABLE V. Comparison of bound-state energies (in cm^{-1}) of He–HCN obtained using the CCSD(T)/avtz+(33221) PES, and 1E8 and 2E8 empirical potentials developed by Atkins and Hutson.[a]

Level (j,L,J)	1E8	2E8	CCSD(T)
(0,0,0)	−9.657 34	−8.888 46	−8.867 29
(1,1,0)	−5.898 97	−5.142 01	−5.138 28
(0,0,0)[b]	−0.463 33	⋯	−0.120 39
(0,1,1)	−9.127 18	−8.358 19	−8.342 74
(1,0,1)	−6.368 29	−5.598 76	−5.554 70
(1,1,1)	−5.861 90	−5.085 85	−5.019 91
(1,2,1)	−4.876 83	−4.108 57	−4.084 34
(0,2,2)	−8.082 28	−7.313 03	−7.308 55
(1,1,2)	−5.743 78	−4.976 50	−4.947 78
(1,2,2)	−4.706 99	−3.939 83	−3.887 20
(1,3,2)	−3.315 28	−2.547 88	−2.522 28
(0,3,3)	−6.555 55	−5.785 59	−5.796 16
(1,2,3)	−4.553 35	−3.786 71	−3.777 81
(1,3,3)	−2.998 73	−2.246 72	−2.213 93
(1,4,3)	−1.254 18	−0.511 90	−0.472 29
(0,4,4)	−4.597 33	−3.825 19	−3.853 09
(1,3,4)	−2.794 61	−2.031 71	−2.051 05
(1,4,4)	−0.770 61	−0.043 38	−0.036 58
(0,5,5)	−2.263 00	−1.486 47	−1.532 96

[a]Reference 16.
[b]Excited van der Waals stretching state $(n=1)$.

energy $(j,L,J)=(1,2,3)\leftarrow(1,2,2)$ transition which is in error by about 30%.

Atkins and Hutson[16] emphasized the differences between their two potentials, but did not consider one of them as superior noting that both reproduced the five observed transitions to better than 1 MHz. Obviously, we think that the 2E8 potential more closely resembles the true PES for He–HCN.

2. Ne–HCN

The agreement between our *ab initio* results and the experimental data for Ne–HCN is much worse than for He–HCN. Table VII contains a list of rotational transition frequencies obtained with the avtz+(33221) PES and the three microwave data of Gutowsky *et al.*[12] The transitions are labeled using both the already introduced (j,L,J) notation as well as the $J_{K_aK_c}$ notation employed for asymmetric tops. Gutowsky *et al.*[12] used the latter notation, but we think that

TABLE VI. Comparison of observed and calculated transition frequencies (in MHz) for He–HCN.

Transition $(j',L',J')\leftarrow(j,L,J)$	2E8[a]	CCSD(T)[b]	Observed[c]
(0,1,1)←(0,0,0)	15 893.7	15 725.7	15 893.6
(0,2,2)←(0,1,1)	31 325.3	31 004.3	31 325.2
(1,1,2)←(0,1,1)	101 432.3	101 778.5	101 432.1
(1,2,3)←(0,2,2)	105 794.9	105 848.8	105 795.3
(1,2,3)←(1,2,2)	4604.4	3279.6	4604.2
(1,0,1)←(0,0,0)	98 662.2	99 309.0	
(1,1,1)←(0,1,1)	98 076.2	99 616.0	
(0,3,3)←(0,2,2)	45 779.2	45 340.3	
(1,3,3)←(0,3,3)	106 173.4	107 392.4	

[a]Reference 16.
[b]This work.
[c]Reference 13.

TABLE VII. Comparison of experimental and calculated rotational transition frequencies (in MHz) for Ne–HCN.

Transition $(j',L',J')\leftarrow(j,L,J)$	Transition $J'_{K'_aK'_b}\leftarrow J_{K_aK_b}$	avtz+(33221)	Observed[a]
(0,1,1)←(0,0,0)	$1_{01}\leftarrow 0_{00}$	6042.4	5540.5
(0,2,2)←(0,1,1)	$2_{02}\leftarrow 1_{01}$	12 066.5	11 050.5
(0,3,3)←(0,2,2)	$3_{03}\leftarrow 2_{02}$	18 052.4	16 500.4
(0,4,4)←(0,3,3)	$4_{04}\leftarrow 3_{03}$	23 977.0	
(0,5,5)←(0,4,4)	$5_{05}\leftarrow 4_{04}$	29 811.7	
(0,6,6)←(0,5,5)	$6_{06}\leftarrow 5_{05}$	35 518.9	

[a]Reference 12.

the former would be more appropriate. This is illustrated in Table VIII in which low-lying vibration–rotation states for $J=0-3$ are grouped into blocks for $j=0-2$. The pattern of energy levels for $j=0$ and $j=1$ is qualitatively similar to that obtained for He–HCN and presented in Table V. A closer examination of the results in Table VIII reveals more significant deviations from case 1 coupling scheme than those found for He–HCN. In particular, L is no longer a nearly good quantum number so it may seem more appropriate to employ case 2 coupling scheme.[41] This, however, is not without pitfalls either, as the pattern of states for $j=1$ and $j=2$ clearly indicates. In case 2 coupling the projection of j onto the intermolecular axis R, denoted by K, is nearly conserved. In the presence of a sufficiently anisotropic potential the energies of pairs of levels for a particular value of $|K|$ should become degenerate. In the case of Ne–HCN they are far from degenerate, especially for $j=2$, as can be seen in Table VIII.

The calculated rotational transition frequencies in Table VII differ from the experimental results by approximately 9%, which is much worse than what we found for He–HCN or for Ar–HCN and Kr–HCN as will be shown later. The possible source of this relatively poor agreement will be discussed in the subsequent section.

In addition to $K=0$ transitions, Gutowsky *et al.*[12] assigned also several $K\neq 0$ transitions, although they noted ''the difficulty in distinguishing *l*-type doublets of a vibrational origin from K doublets for rotation of an asymmetric top.''[12] The pattern of vibration–rotation energies in Table VIII is quite different from that observed for other weak intermolecular complexes such as Ar–CO,[21] which behave as near asymmetric tops. As a result, the identification of energy levels involved in transitions assigned by Gutowsky *et al.* is far from obvious. We made the most straightforward choices, but it is possible that the actual transitions could occur between states different from those that we chose. The two observed transitions assigned as $2_{12}\leftarrow 1_{11}$ and $2_{11}\leftarrow 1_{10}$ in case 1 coupling notation could be written as (1,1,2)←(1,0,1) and (1,2,2)←(1,1,1). The agreement is very poor as the experimental transition frequencies are 11 011 and 11 085 MHz, whereas we find 9922 and 12 664 MHz. The large differences between the experimental data and theoretical predictions are the most noticeable, but we would like to draw attention to the more qualitative issue. The two measured transitions have very similar frequencies indicative

TABLE VIII. Energies (in cm^{-1}) of several vibration–rotation states of Ne–HCN. For a given J the levels are labeled (j,L).

$J=0$		$J=1$		$J=2$		$J=3$	
				(2,4)	−27.463	(2,5)	−26.605
				(2,3)	−27.727	(2,4)	−27.019
		(2,3)	−28.094	(2,2)	−29.172	(2,3)	−28.598
		(2,2)	−28.206	(2,1)	−29.451	(2,2)	−28.937
(2,2)	−29.534	(2,1)	−29.462	(2,0)	−29.523	(2,1)	−29.180
		(1,2)	−33.917	(1,3)	−33.467	(1,4)	−32.800
		(1,1)	−35.219	(1,2)	−34.796	(1,3)	−34.163
(1,1)	−34.144	(1,0)	−35.267	(1,1)	−34.936	(1,2)	−34.432
(0,0)	−37.942	(0,1)	−37.741	(0,2)	−37.338	(0,3)	−36.736

of a complex with large anisotropy. On the other hand, the calculated transition frequencies suggest a complex with small anisotropy. To explain this discrepancy, we examined the trends for all four Rg–HCN complexes using the data in Tables I–IV. As expected, the anisotropy gradually increases from He–HCN to Kr–HCN, although the results for Ne–HCN obtained with the avtz+(33221) basis set clearly deviate from the trend. This problem will be discussed in more detail later in this work. We note that Gutowsky *et al.* performed similar analysis and found the PES for Ne–HCN to be more isotropic than the surfaces for either Ar–HCN or Kr–HCN. Overall, it appears quite likely that $K\neq 0$ transitions observed by Gutowsky *et al.*[12] were misassigned. Further experimental study of Ne–HCN with measurements for higher J would certainly be very useful.

3. Ar–HCN

A comparison of experimental and theoretical results for Ar–HCN and Ar–DCN is presented in Table IX. The vibration–rotation states are labeled using case 2 coupling scheme[41] and the notation that was developed to describe complexes such as Ar–HCl.[42,43] The Greek letters Σ and Π denote states with $K=0$ and $|K|=1$, and subscripts 0 or 1 denote the value of j to which a given state can be related. The ground state is thus denoted by Σ_0 and the excited bending states by Σ_1 and Π_1. In addition, the parity label, $p=(-1)^{j+L+J}$, is used to differentiate between levels with $p=+1$ and $p=-1$ which are designated by superscripts *e* and *f*, respectively.

The notation employed in Table IX can be easily con-

TABLE IX. Comparison of experimental and calculated energies (in cm^{-1}) of vibration–rotation states of Ar–HCN and Ar–DCN.

		Ar–HCN				Ar–DCN	
State	J	MP4[a]	CCSD(T) avtz+(332)[c]	CCSD(T) avtz+(33221)[d]	Observed[b]	CCSD(T) avtz+(33221)[d]	Observed[b]
Σ_0	1	0.1053	0.1112	0.1071	0.1074	0.1050	0.1050
	2	0.3156	0.3335	0.3213	0.3220	0.3148	0.3151
	3	0.6307		0.6422	0.6436	0.6293	0.6299
	4	1.0502		1.0695	1.0718	1.0483	1.0493
	5	1.5736		1.6028	1.6061	1.5714	1.5730
Σ_1	0	5.0496	4.2391	5.7694	5.5002	6.2632	6.3049
	1	5.1735	4.3644	5.8902	5.6264	6.3322	6.4154
	2	5.4216	4.6153	6.1338	5.8796	6.5449	6.6453
	3	5.7952		6.5028	6.2608	6.8869	7.0003
	4	6.2954		6.9993	6.7714	7.3574	7.4831
	5	6.9232		7.6249	7.4124	7.9562	8.0943
Π_1^f	1	5.7977	4.8567	6.2363	6.1381	6.3948	6.5885
	2	6.0633	5.1241	6.5034	6.4090	6.6540	6.8509
	3	6.4616		6.9038	6.8151	7.0425	7.2442
	4	6.9923		7.4370	7.3559	7.5597	7.7678
	5	7.6549		8.1027	8.0310	8.2052	8.4211
Π_1^e	1	5.8009		6.2447	6.1426	6.4526	6.6066
	2	6.0726		6.5266	6.4217	6.7530	6.8963
	3	6.4787		6.9457	6.8389	7.1801	7.3201
	4	7.0181		7.4996	7.3924	7.7345	7.8753
	5	7.6895		8.1865	8.0810	8.4159	8.5603

[a]Reference 14.
[b]Calculated from molecular constants given in Ref. 20.
[c]Reference 19.
[d]This work.

TABLE X. Comparison of experimental and calculated rotational transition frequencies (in MHz) for Kr–HCN and Kr–DCN.

	Kr–HCN		Kr–DCN	
Transition ($J'\leftarrow J$)	CCSD(T) avtz+(33221)	Observed[a]	CCSD(T) avtz+(33221)	Observed[a]
1←0	2321.41	2362.92	2284.84	2317.18
2←1	4642.32	4724.72	4569.35	4633.73
3←2	6962.23	7084.35	6853.08	6949.00
4←3	9280.65	9440.82	9135.73	9262.43
5←4	11 597.14	11 793.33	11 416.94	11 573.49

[a]Reference 1.

verted to the one used earlier for He–HCN and Ne–HCN. For example, the Σ_1, Π_1^f, and Π_1^e states for $J=1$ can be labeled (1,0,1), (1,1,1), and (1,2,1). Most studies concerned with intermolecular complexes are focused on a single system so it is not necessary to switch between various coupling and labeling schemes, although, as noted by Hutson,[41] different coupling cases can be observed for different states of the same intermolecular complex. Here a problem of labeling arose because of the choices made by researchers who studied a given complex earlier. While one of the labeling schemes may be more appropriate in a given case, it is important to be able to switch between them when discussing related complexes. It is also important to realize which quantum numbers are rigorously good and which are not. In the present case only J and p are good quantum numbers, while j, L, and K are not.

The energies of the ground state rotational levels obtained from the avtz+(33221) potential are in excellent agreement with the spectroscopic data.[2,3,5] The calculated energies are in error by no more than 0.2% for Ar–HCN and 0.1% for Ar–DCN. The energies of the excited Σ_1 and Π_1 states which are usually more difficult to reproduce are also in good agreement with the experimental results. For the Σ_1 state of Ar–HCN the differences between the calculated and experimental energies range from 4.9% for $J=0$ to 2.9% for $J=5$. The values of energies for both of Π_1 states are somewhat closer to their experimental counterparts as the error varies from 1.7% to 0.9%. Slightly better agreement with the experimental energies is obtained for the Σ_1 and Π_1 states of Ar–DCN.

Table IX contains also vibration–rotation energies reported in two previous *ab initio* studies.[14,19] As can be seen, our results represent a significant improvement over both of them.

4. Kr–HCN

After analyzing the results for the three lighter Rg–HCN complexes and finding very unequal degree of success in reproducing the experimental results, we were very pleased to discover that our approach performed surprisingly well for Kr–HCN. Because of the size of the Kr atom we expected that Kr–HCN would be more difficult to study than other members of the Rg–HCN family. Fortunately, our concerns turned out to be unwarranted and, as can be seen in Table X, we were able to obtain very good agreement with the experimental results of Campbell *et al.*,[1] who reported rotational spectra of several isotopomers of Kr–HCN. In Table X we compare the transition frequencies for two of these isotopomers: ^{84}Kr–HCN and ^{84}Kr–DCN. The differences between the calculated and experimental data do not exceed 1.8%.

Good agreement with microwave spectra notwithstanding, we think that it is premature to declare that the PES for Kr–HCN is comparable in quality to the He–HCN or Ar–HCN potentials. A more meaningful assessment of the overall quality of the avtz+(33221) surface for Kr–HCN could be made if the transition frequencies for Σ_1 and Π_1 bending states were available. We present our predictions in the following section.

IV. DISCUSSION

A. Reliability of *ab initio* potentials

In several previous studies[21–24] in addition to the CCSD(T) calculations with the avtz+(33221) basis set we also performed calculations for selected configurations in the vicinity of the minima using a larger avqz+(33221) basis set. For most complexes the results obtained with the two basis sets were very similar. The biggest difference so far has been found for the $\tilde{A}\,^2\Sigma^+$ state of the Ne·SH complex for which the depth of one of the wells calculated with the avqz+(33221) basis set was 6.5 cm^{-1} (30 μE_h) deeper than that obtained with the avtz+(33221) basis set.[23] This difference amounts to 6.3% of the avtz+(33221) value and seems to imply that large differences might be seen in transition frequencies. We verified that this was not the case by developing the entire PES using the avqz+(33221) basis set and performing subsequent variational calculations of vibration–rotation energy levels. The calculated transition frequencies were found to be very similar to those obtained with the avtz+(33221) PES.[23]

Despite the outcome of this test we think that it is important to investigate how basis set size affects both the interaction energies and rovibrational eigenstates. It is not yet clear whether the case of the $\tilde{A}\,^2\Sigma^+$ state of Ne·SH is exceptional or not, and because of that we intend to continue to perform calculations with basis sets of different size.

In the present case we focused on the anisotropy of each surface which we defined as the difference between the energy of the global minimum at $\theta=0°$, and the highest energy saddle point on the minimum energy path which, as can be seen in Fig. 1, in every case is either at or very near $\theta=180°$. We note that the just defined anisotropy is a much more sensitive function of basis set size than the interaction energies from which it is evaluated. The anisotropies of He–HCN, Ne–HCN, Ar–HCN, and Kr–HCN PESs obtained with the avtz+(33221) basis set are, respectively, 46.5, 37.2, 170.0, and 224.5 μE_h. The unexpectedly small anisotropy of the Ne–HCN PES is apparent, making additional calculations with the avqz+(33221) basis set an obvious necessity. The results of these calculations are as follows: the PESs for He–HCN, Ne–HCN, and Ar–HCN become more anisotropic by approximately 5%, 95%, and less than 1%, respectively, but the PES for Kr–HCN becomes less anisotropic by approximately 12%. The small changes of the anisotropies of the avtz+(33221) surfaces for He–HCN, Ar–HCN, and to a lesser degree also for Kr–HCN clearly show that our treatment is adequate and that the results of calculations with larger basis sets are not expected to be significantly different from ours. On the other hand, the enormous change in the anisotropy of the avtz+(33221) PES for Ne–HCN reveals an evident deficiency of the potential. Interestingly, the differences between the interaction energies obtained with the two basis sets for configurations along the minimum energy path vary considerably for different values of θ. The greatest differences are found in the neighborhoods of linear configurations. When the avqz+(33221) basis set is used the depth of the $\theta=0°$ well increases by approximately 20 μE_h and the depth of the $\theta=180°$ well decreases by approximately 15 μE_h. The differences for nonlinear configurations vary monotonically between these two extrema.

Besides the difference in anisotropy, there are also important qualitative differences between the avtz+(33221) and avqz+(33221) PESs for Ne–HCN. While the former contains four minima, the latter contains only two with the global minimum in the linear Ne–HCN configuration at $R=8.00\,a_0$ and a local one at $R=6.62\,a_0$ and $\theta=102.0°$. Their depths are 61.56 cm^{-1} ($280.5\,\mu E_h$) and 52.48 cm^{-1} ($239.1\,\mu E_h$), respectively. Importantly, the linear Ne–NCH configuration on the avqz+(33221) surface represents a saddle point in contrast to the avtz+(33221) potential where it is a local minimum.

The calculated rotational transition frequencies obtained with the avqz+(33221) PES are in much better agreement with the experimental results than those reported in Table VII. The three calculated transitions at low J differ from their experimental counterparts[12] by approximately 1.3% and have the following frequencies: 5612, 11 194, and 16 717 MHz. Because of the greater anisotropy of the avqz+(33221) PES the frequencies of the calculated rovibrational (1,1,2)←(1,0,1) and (1,2,2)←(1,1,1) transitions do not differ as much as those obtained with the avtz+(33221) PES. Nonetheless, they are still much further apart than the microwave transitions assigned by Gutowsky *et al.*[12]

Despite some quantitative differences between the energies of vibration–rotation states of Ne–HCN obtained using the two surfaces, they share many qualitative similarities. The most important of them is the pattern of energy levels for $j=1$. The (1,0,1) and (1,1,1) levels have similar energies and if case 2 coupling is used they can be labeled as Π_1 ($j=1$, $K=1$). The (1,1,0) level which, as can be seen in Table VIII, is higher in energy than Π_1 states, can be labeled as Σ_1 ($j=1$, $K=0$). Thus the energy progression for Ne–HCN is qualitatively different from that for Ar–HCN shown in Table IX. The relative energies of the Σ_1 and Π_1 states have been shown to depend on the sign of the coefficient proportional to the $P_2(\cos\theta)$ term in the intermolecular potential.[6,41] For Ne–HCN as well as Ar–HCN this coefficient should be negative since in both cases the global minima are in the linear $\theta=0°$ configurations. Perhaps the observed difference in ordering of the Σ_1 and Π_1 states is caused by other terms in the intermolecular potentials of Ne–HCN and Ar–HCN, but this will have to be verified. We hope that this will be done in not too distant future as this problem should be of interest to both theorists and experimentalists.

We end this section by noting that while we have compared the results obtained with two different basis sets, we have not yet compared the results obtained by using different methods. The CCSD(T) approach is not the most sophisticated *ab initio* approach that has been developed so far, but, at least for the moment, it represents the highest level of theory that can be used without compromising basis set quality to calculate potential energy surfaces of complexes such as those considered in this work. The use of more sophisticated approaches like, for example, the full single, double, and triple excitation coupled-cluster (CCSDT) method[44] is still too expensive. We have compared our CCSD(T) results to the results obtained using second-order Møller–Plesset perturbation theory (MP2) and the single and double excitation coupled cluster approach (CCSD), both of which are byproducts of CCSD(T) calculations. With only one exception the transition frequencies obtained using the CCSD(T) PESs are in much better agreement with the experimental data than either the CCSD or MP2 results. Only in the case of rotational transition frequencies for Ne–HCN calculated using the avtz+(33221) PESs the CCSD results are closer to the experimental values than their CCSD(T) counterparts.

B. Spectroscopic predictions

In addition to the results that we compared with the available experimental data in Sec. III, we have also found values of many quantities that have not yet been measured or have not been measured for all four complexes. These results are presented in this section.

We begin with Table XI which contains molecular constants for the ground states of the four complexes and their deuterated analogs. Even though some of the results in Table XI have been presented in a different form in earlier tables, we decided to collect data for all complexes in a single table to better illustrate how the increasing anisotropy of the intermolecular potential influences the spacing of rotational levels. The rotational transition frequencies through $J=9$ have been fitted to the equation

$$E(J)=BJ(J+1)-D_JJ^2(J+1)^2+H_JJ^3(J+1)^3+L_JJ^4(J+1)^4+P_JJ^5(J+1)^5. \tag{8}$$

For Kr–HCN and Kr–DCN only three constants were needed to obtain a fit of reasonable quality. To fit five rotational transition frequencies that are possible between six bound $j=0$ states of He–HCN and He–DCN, we needed four constants because of unusually large centrifugal distortion effects. In the case of He–DCN this was still not entirely sufficient as the quality of the fit was the worst of all reported in Table XI. The centrifugal distortion even in Kr–HCN is considered to be unusually large,[1,10] although it is much smaller than in complexes of lighter rare gases. As can be seen, the magnitude of the D_J centrifugal distortion con-

TABLE XI. Spectroscopic constants obtained by fitting the ground state rotational energies, as described in the text.

Complex/PES	B (MHz)	D_J (MHz)	H_J (kHz)	L_J (Hz)	P_J (mHz)
He–HCN/avtz+(33221)	7898.89	17.79	−122.7	2757	
He–HCN/2E8[a]	7985.58	18.25	−140.3	3151	
He–DCN/avtz+(33221)	7692.32	25.32	−20.55	3822	
Ne–HCN/avtz+(33221)	3022.76	0.7560	−1.543	−14.94	57.85
Ne–HCN/avqz+(33221)	2808.69	1.260	0.1848	12.73	−36.83
Ne–DCN/avtz+(33221)	2964.72	0.8810	−2.146	−22.01	141.7
Ne–DCN/avqz+(33221)	2732.60	1.210	1.300	7.876	−33.23
Ar–HCN/avtz+(33221)	1606.28	0.1627	0.3028	−0.4734	0.1678
Ar–HCN/exp[b]	1609.84	0.1731	0.3711	−0.8117	1.384
Ar–DCN/avtz+(33221)	1573.47	0.1085	0.1678	−0.1458	−0.5217
Ar–DCN/exp[b]	1574.79	0.1020	0.1749	−0.3552	0.6132
Kr–HCN/avtz+(33221)	1160.75	0.021 07	0.012 06		
Kr–HCN/exp[c]	1181.55	0.046 91	0.066 60		
Kr–DCN/avtz+(33221)	1142.46	0.015 51	0.006 64		
Kr–DCN/exp[c]	1158.65	0.026 97	0.027 20		

[a]Reference 16.
[b]Reference 20.
[c]Reference 1.

stant increases by approximately an order-of-magnitude for each lighter rare gas atom.

In Tables XII–XIV we present the energies and expectation values of several quantities for the ground, (1,1,0), and (1,1,1) states. The latter two can alternatively be denoted as Σ_1, $J=0$ and Π_1, $J=1$, respectively. The calculated values of $\langle R\rangle$ in the ground state of each complex can be compared with the values obtained from the observed rotational constants and nuclear quadrupole coupling constants.[2,3,10,12,13] The agreement is good, but it can be improved by calculating $\langle R^{-2}\rangle^{-1/2}$ instead of $\langle R\rangle$. Drucker *et al.*[13] noted that $\langle R\rangle$ is significantly increased in He–HCN and decreased in Ne–HCN. The results in Table XII confirm this observation, but when considering all four complexes the large value of $\langle R\rangle$ in He–HCN appears as much more unusual.

Further irregularities in the average intermolecular distance can be seen by comparing the results in Tables XII and XIII. $\langle R\rangle$ in both complexes of Ar and Kr decreases very significantly in the excited state, but for He–HCN, He–DCN, and Ne–HCN the opposite is true. The value of $\langle R\rangle$ in Ne–DCN is found to be virtually the same for both the ground and the excited state. The calculated values of $\langle R\rangle$ for all complexes in their (1,1,1) states decrease in comparison to the corresponding ground state values. The standard deviation for the intermolecular distance is very large for the helium complexes indicating extensive radial motions. As expected, $(\langle R^2\rangle-\langle R\rangle^2)^{1/2}$ decreases for complexes of heavier rare gas atoms.

The calculated values of $\langle P_1(\cos\theta)\rangle$ for the three states can be compared with the values given by Drucker *et al.*[11] for Ar–HCN. The agreement is not good, but this is not surprising because of the approximate nature of formulas that are used to find $\langle P_1(\cos\theta)\rangle$ from the experimental results. Much better agreement is typically seen between the calculated and experimental values of $\langle P_2(\cos\theta)\rangle$. This is indeed the case for Ar–HCN and Kr–HCN. Good agreement between our and the experimental[13] $\langle P_2(\cos\theta)\rangle$ values for He–HCN appears to be accidental, because, as explained by Atkins and Hutson,[16] for complexes of light atoms and molecules with large moments of inertia it is no longer adequate

TABLE XII. Calculated spectroscopic properties of the indicated complexes in their ground states.

	Energy (cm^{-1})	$\langle R\rangle$ (Å)	$(\langle R^2\rangle-\langle R\rangle^2)^{1/2}$ (Å)	$\langle P_1(\cos\theta)\rangle$	$\langle P_2(\cos\theta)\rangle$
He–HCN	−8.867	4.328	0.582	0.347	0.088
He–DCN	−9.001	4.333	0.579	0.405	0.118
Ne–HCN[a]	−37.942	3.836	0.303	0.157	−0.070
Ne–HCN[b]	−38.854	3.950	0.321	0.501	0.146
Ne–DCN[a]	−38.327	3.834	0.301	0.236	−0.050
Ne–DCN[b]	−39.058	3.926	0.310	0.517	0.156
Ar–HCN	−111.096	4.375	0.249	0.820	0.579
Ar–DCN	−112.731	4.353	0.223	0.851	0.635
Kr–HCN	−140.136	4.573	0.189	0.893	0.720
Kr–DCN	−142.247	4.535	0.151	0.908	0.752

[a]Calculated using the avtz+(33221) potential.
[b]Calculated using the avqz+(33221) potential.

TABLE XIII. Calculated spectroscopic properties of the indicated complexes in their $(j,L,J)=(1,1,0)$ (or Σ_1) states.

	Energy (cm^{-1})	$\langle R\rangle$ (Å)	$(\langle R^2\rangle-\langle R\rangle^2)^{1/2}$ (Å)	$\langle P_1(\cos\theta)\rangle$	$\langle P_2(\cos\theta)\rangle$
He–HCN	−5.138	4.441	0.603	−0.169	0.333
He–DCN	−5.702	4.426	0.604	−0.197	0.306
Ne–HCN[a]	−34.144	4.035	0.312	0.114	0.361
Ne–HCN[b]	−34.611	3.942	0.355	0.082	0.172
Ne–DCN[a]	−34.964	4.008	0.305	0.082	0.325
Ne–DCN[b]	−34.550	3.930	0.339	0.071	0.158
Ar–HCN	−105.326	3.978	0.290	0.190	−0.149
Ar–DCN	−106.468	3.957	0.268	0.234	−0.166
Kr–HCN	−131.003	4.072	0.236	0.216	−0.213
Kr–DCN	−132.301	4.065	0.225	0.276	−0.199

[a]Calculated using the avtz+(33221) potential.
[b]Calculated using the avqz+(33221) potential.

TABLE XIV. Calculated spectroscopic properties of the indicated complexes in their $(j,L,J)=(1,1,1)$ (or Π_1^f) states.

	Energy (cm^{-1})	$\langle R\rangle$ (Å)	$(\langle R^2\rangle-\langle R\rangle^2)^{1/2}$ (Å)	$\langle P_1(\cos\theta)\rangle$	$\langle P_2(\cos\theta)\rangle$
He–HCN	−5.016	4.227	0.589	0.097	−0.184
He–DCN	−5.627	4.225	0.586	0.118	−0.177
Ne–HCN[a]	−35.219	3.753	0.270	0.047	−0.235
Ne–HCN[b]	−35.353	3.768	0.275	0.175	−0.201
Ne–DCN[a]	−36.016	3.748	0.266	0.081	−0.231
Ne–DCN[b]	−35.469	3.761	0.272	0.193	−0.195
Ar–HCN	−104.859	3.966	0.228	0.266	−0.186
Ar–DCN	−106.337	3.976	0.229	0.342	−0.134
Kr–HCN	−130.156	4.120	0.229	0.329	−0.137
Kr–DCN	−131.981	4.146	0.234	0.429	−0.046

[a]Calculated using the avtz+(33221) potential.
[b]Calculated using the avqz+(33221) potential.

TABLE XVI. The predicted energies (in cm^{-1}) of the ten lowest bending and stretching states of Ar–HCN and Ar–DCN. All levels given are for $J=0$.

Ar–HCN		Ar–DCN	
Label (nb)	E(nb)	Label (nb)	E(nb)
00	0.000	00	0.000
01	5.769	01	6.263
02	14.190	02	14.031
03	21.008	03	20.436
04	28.839	04	27.086
10	36.453	10	35.118
11	41.406	11	38.478
12	42.215	05	41.319
05	50.499	12	46.597
13	51.123	13	50.228

to assume that the component of nuclear quadrupole coupling constant along the inertial a-axis of the complex is given by the equation

$$(eqQ)_{aa}=(eqQ)_{\text{monomer}}\langle P_2(\cos\theta)\rangle. \tag{9}$$

It is not clear to what extent this applies to the neon complexes, but perhaps it is one of the reasons responsible for the lack of agreement between the calculated and the experimental values of $(eqQ)_{aa}$ for Ne–HCN.[12] We note, however, that for Ne–DCN the agreement between theory and experiment is better than for Ne–HCN.

In Tables XV–XVII we present the energies of the ten lowest $J=0$ rotation–vibration states for the neon, argon, and krypton complexes. To label the states, we use two quantum numbers: an intermolecular stretching quantum number n and a bending quantum number b, which correlates in the isotropic limit with the HCN rotational quantum number. Neither n nor b are good quantum numbers, but they are useful as labels. Details of our procedure for assigning the (nb) labels will be given in a forthcoming work.[45] Because of the small values of rotational constants of HCN and DCN several lowest energy states correlate in the isotropic limit with the rotational states of the appropriate molecule. The calculated stretching frequencies of Ne–HCN and Ar–HCN are very similar to the stretching frequencies of Ne–HCl (Ref. 46) and Ar–HCl.[47] The difference in the values of stretching frequencies of Kr–HCN and Kr–DCN simply reflects the limited accuracy of our potential. A small increase of the depth of the global minimum could result in a different assignment in which positions of (10) and (04) states are switched.

Finally, in Table XVIII we give the energies of states of Ar–HCN and Kr–HCN which in the isotropic limit correlate with $j=1$ or $j=2$ rotational states of HCN. The states are labeled using case 1 coupling scheme, but of course it is possible to use case 2 coupling notation and denote the appropriate states as Σ_2, Π_2, and Δ_2. The energies in Table XVIII can be compared with those in Table VIII which contains the results for Ne–HCN. We note that the energies of $j=0$ and $j=1$ states of Ar–HCN are presented in Table IX and $j=0$ states of Kr–HCN in Table X. As can be seen, the energy of the (1,1,1) state of Kr–HCN is higher than the energy of the (1,1,0) state, so the frequency of the Σ_1 transition should be smaller than the frequency of the Π_1 transition. This is qualitatively similar to what has been found for Ar–HCN, although the frequencies are higher as predicted values are 9.1 and 10.0 cm^{-1}. Surprisingly, the reversed order is expected for the Σ_2 and Π_2 states of both Ar–HCN and Kr–HCN. The former state is predicted to be 14.2 cm^{-1} and the latter one 13.5 cm^{-1} higher than the ground state of

TABLE XV. The predicted energies (in cm^{-1}) of the ten lowest bending and stretching states of Ne–HCN and He–DCN. All levels given are for $J=0$.

Ne–HCN			Ne–DCN		
Label (nb)	E(nb)[a]	E(nb)[b]	Label (nb)	E(nb)[a]	E(nb)[b]
00	0.000	0.000	00	0.000	0.000
01	3.798	4.242	01	3.363	4.508
02	8.408	9.558	02	7.415	9.750
03	15.883	16.741	03	13.766	16.975
10	22.694	23.656	10	21.321	23.741
11	24.171	24.783	11	22.879	24.940
12	27.497	28.234	04	26.437	28.484
04	32.333	33.154	12	29.979	33.376
20	33.195	34.043	20	30.630	34.250
21	36.073	36.828	13	34.995	37.013

[a]Calculated using the avtz+(33221) potential.
[b]Calculated using the avqz+(33221) potential.

TABLE XVII. The predicted energies (in cm^{-1}) of the ten lowest bending and stretching states of Kr–HCN and Kr–DCN. All levels given are for $J=0$.

Kr–HCN		Kr–DCN	
Label (nb)	E(nb)	Label (nb)	E(nb)
00	0.000	00	0.000
01	9.134	01	9.946
02	17.820	02	18.044
03	24.532	03	24.400
10	31.712	04	30.981
04	37.530	10	36.291
11	43.449	11	40.974
12	46.093	12	45.115
05	52.420	05	49.002
13	56.189	13	55.137

TABLE XVIII. Energies (in cm^{-1}) of the indicated vibration–rotation states of Ar–HCN and Kr–HCN. For a given J the levels are labeled (j,L).

	$J=0$		$J=1$		$J=2$		$J=3$	
Ar–HCN					(2,4)	−96.433	(2,5)	−95.971
					(2,3)	−97.352	(2,4)	−96.940
			(2,3)	−96.747	(2,2)	−97.439	(2,3)	−97.101
			(2,2)	−97.626	(2,1)	−98.912	(2,2)	−98.542
	(2,2)	−96.906	(2,1)	−97.657	(2,0)	−98.913	(2,1)	−98.545
Kr–HCN					(2,4)	−122.011	(2,5)	−121.706
					(2,3)	−123.604	(2,4)	−123.281
			(2,3)	−122.214	(2,2)	−123.624	(2,3)	−123.318
			(2,2)	−123.822	(2,1)	−124.061	(2,2)	−123.809
	(2,2)	−122.316	(2,1)	−123.829	(2,0)	−124.062	(2,1)	−123.814
			(1,2)	−130.156	(1,3)	−129.961	(1,4)	−129.669
			(1,1)	−130.156	(1,2)	−129.961	(1,3)	−129.668
	(1,1)	−131.003	(1,0)	−130.905	(1,1)	−130.709	(1,2)	−130.416

Ar–HCN. For Kr–HCN, the analogous values are 17.8 and 16.3 cm^{-1}. These results are very unexpected as the intermolecular potential is thought to affect different van der Waals bending states in a qualitatively similar way.[6,41] A confirmation of this finding or its dismissal as a computational artifact will have to wait until the appropriate transitions are observed.

V. SUMMARY

The two-dimensional PESs for the He–HCN, Ne–HCN, Ar–HCN, and Kr–HCN complexes have been calculated using the CCSD(T) approach and the augmented correlation consistent polarized basis sets with an additional $(3s3p2d2f1g)$ set of bond functions.[48] The global minima are in the linear Rg–HCN configurations with well depth ranging from 29.90 cm^{-1} for He–HCN to 179.0 cm^{-1} for Kr–HCN. The surfaces contain also local minima at bent geometries. For selected geometries of He–HCN, Ar–HCN, and Kr–HCN we have performed additional calculations using the avqz+(33221) basis sets and have found only small differences in comparison with the results obtained using the avtz+(33221) basis sets. For Ne–HCN, on the other hand, the differences between the avtz+(33221) and avqz +(33221) surfaces have been found to be much more pronounced and that has forced us to develop the entire PES using the latter basis set.

The results of calculations of the bound vibration–rotation states of He–HCN, Ar–HCN, and Kr–HCN are in very good agreement with the available experimental results. Unfortunately, the same cannot be said about the results for Ne–HCN obtained using the avtz+(33221) PES, as the calculated and experimental rotational transition frequencies differ by 9%. The differences decrease to below 2% if the avqz+(33221) PES is used.

In addition to calculating the frequencies of transitions that have already been measured, we have also used the newly developed PESs to predict the frequencies of transitions that have yet to be measured. The Σ_1 and Π_1 transitions for Kr–HCN are predicted to have frequencies of 9.1 and 10.0 cm^{-1}. A very interesting prediction is that the Π_2 states of both Ar–HCN and Kr–HCN should be lower in energy than the Σ_2 states. Observation of these states would provide a very stringent test of the quality of our potentials.

We note that only very recently it has become possible to calculate PESs that reproduce most and in some cases all experimental results with an error smaller than 5%. The results in Table IX can serve as a good example of progress that has been achieved. Obviously, the quality of *ab initio* PESs is judged primarily by how well they can reproduce the existing experimental data. In cases where agreement with experiment is not satisfactory, it is possible to develop more accurate surfaces by using larger or modified basis sets. This and two previous studies[24,21] demonstrate this clearly. Nonetheless, for complexes such as Ne–HCN for which only small amount of experimental data is available, there is no guarantee that PES obtained with a modified basis set will be better in predicting spectroscopic observables.

The vast majority of *ab initio* studies of weak intermolecular complexes are concerned with only a single complex and in particular with the accuracy with which the experimental results can be reproduced. In this work we have examined a group of four related complexes, and that allowed us to see certain limitations of the presently available *ab initio* methods. For example, it is not yet valid to assume that by performing *ab initio* calculations at a given level of theory with a given basis set one can reach a desired level of accuracy. In this study we have demonstrated that the use of the same method, CCSD(T), and the same basis set, avtz +(33221), resulted in very good agreement with the experimental data for three out of four complexes examined. In the case in which our approach turned out to be deficient, we have shown that the avtz+(33221) basis set was the reason for this deficiency. By using a larger avqz+(33221) basis set we have obtained much better agreement with the experimental results. Unfortunately, the size of this basis set will limit its use to complexes not much larger than those studied here.

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